

United States Energy Consumption Analysis and Modeling

Harbaksh Kaur

Department of Atmospheric and Oceanic Science, University of California Los Angeles

AOS C111: Introduction to Machine Learning for Physical Science

Introduction

Energy consumption is a pressing topic globally as most of the world's energy comes from fossil fuels and rising consumption levels means a negative impact on the environment. More energy consumption will contribute to climate change from the greenhouse gasses released from the production process. Energy is essential for an industrialized society as it powers entire industries, cities, and homes. Energy is used for air conditioning and heating, powering medical equipment, powering our electronics, machinery, and even public transportation systems. According to the Environmental Protection Agency, in 2022 25% of the United States's greenhouse gas emissions came from electricity use. Understanding energy consumption patterns for electricity usage would be beneficial for policymakers to create policy towards shifting to renewable energy.

Data

The data will be provided by a Kraggle data set that looks at global energy consumption trends from 2000-2020.

<https://www.kaggle.com/datasets/anshtanwar/global-data-on-sustainable-energy>

The key variables this report will be focusing on are:

- Electricity from fossil fuels (TWh): Electricity generated from fossil fuels in terawatt hours.
- Electricity from renewables (TWh): Electricity generated from renewable sources in terawatt hours.
- Primary energy consumption per capita (kWh/person): Energy consumption per person in kilowatt-hours.

The pre-processing steps included removing all invalid data values or missing values as well as removing the data of any NaN values. The dataset was further narrowed by focusing on the United States.

Modeling

The modeling for this project focused on finding the best machine learning algorithm that provides the best fit for its ability to show the relationship between renewable and nonrenewable

energy as well as train a linear regression model that would accurately predict energy consumption levels from renewable and nonrenewable energy,

I first isolated the variable energy consumption per person, the target variable, and normalized the data. The data was then split with a `train_test_split` function with a 20% test size and a random state of 42. a linear regression model to determine fit and the results yielded a 95.4% accuracy rate which is very high meaning a linear regression model would provide an accurate fit. The RSME was 1478.95 meaning the model will be off around 1479 kWh/person which is reasonable given that the average American's energy consumption ranges from 70,000 kWh/person to 90,000 kWh/person. While the model fits the data well, there are some outliers once the energy consumption gets very high to levels of around 90,000 kWh/person, however, it is unlikely that it would impact the model severely due to the high accuracy rate. To further check the accuracy and fit of the linear regression model, I also plotted the residuals.

Results

Figure 1. Linear Regression of Expected Energy Consumption (kWh/person)

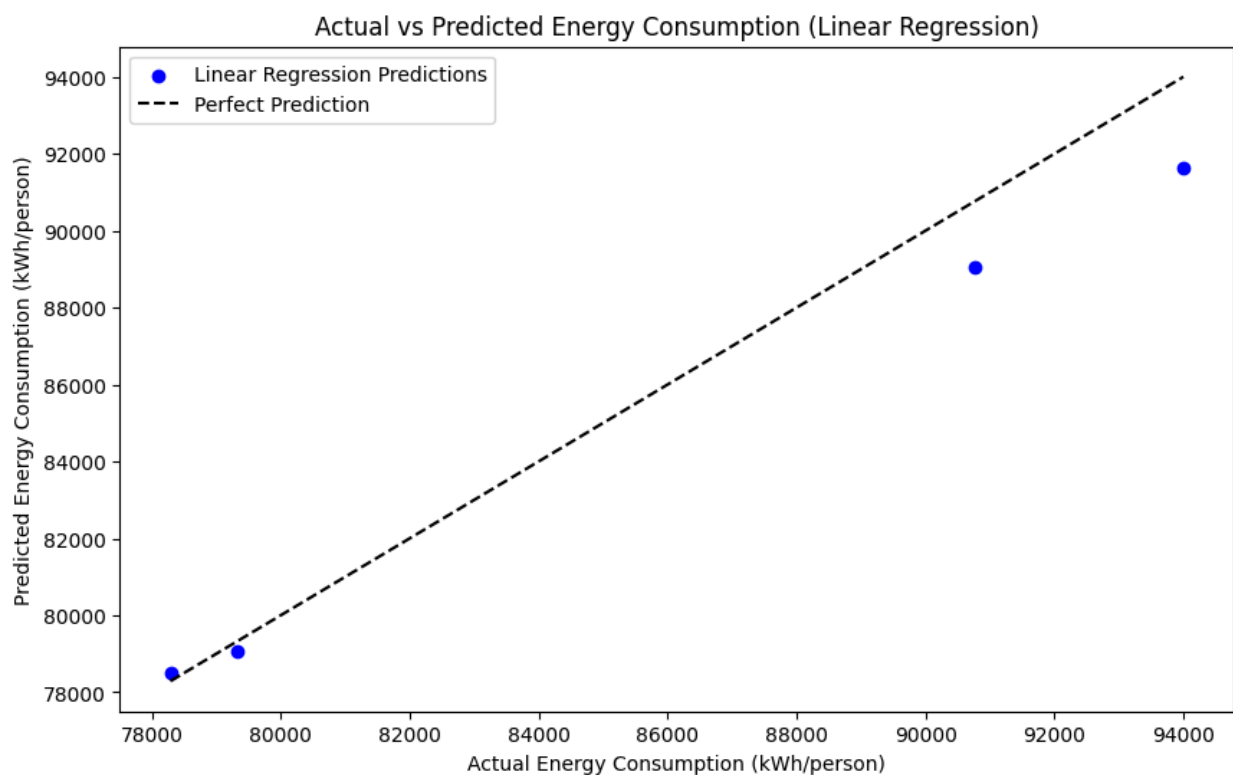


Figure 2. Residuals Plot

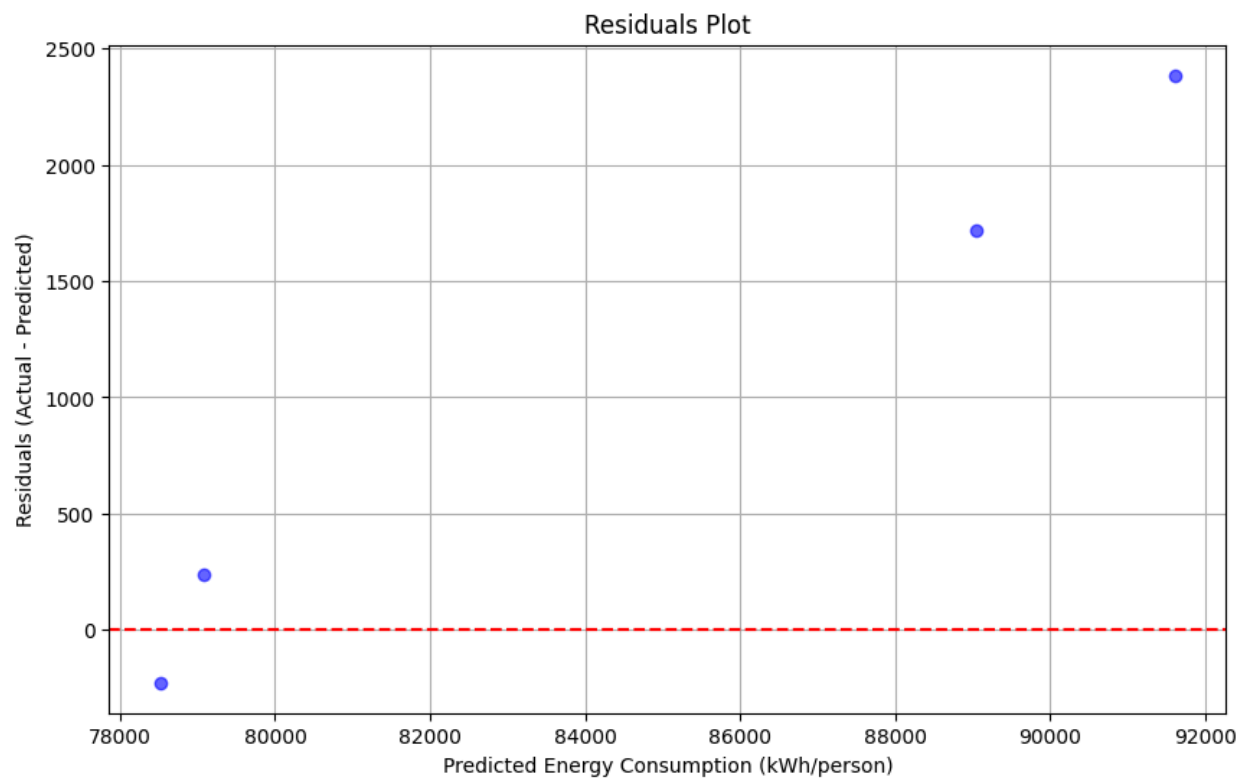


Figure 3. Decision Tree Model for Renewable vs. Fossil Fuel

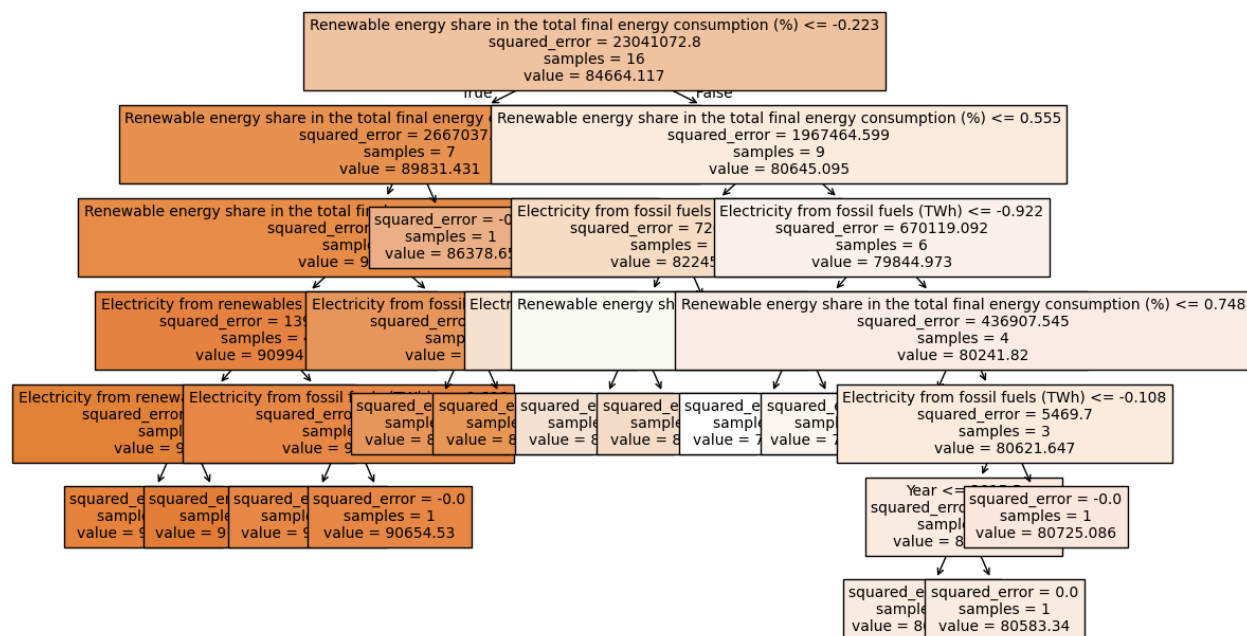


Figure 4. Scatter Plot for Fossil Fuels vs. Renewable Sources

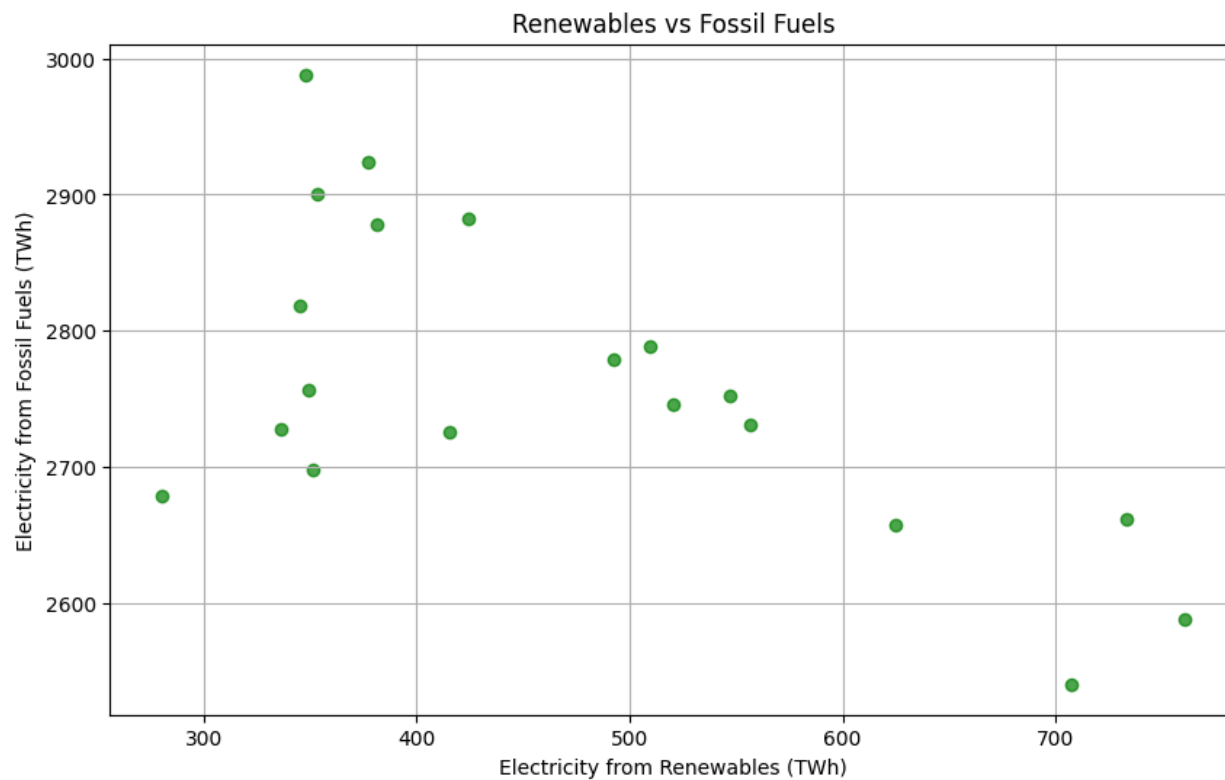
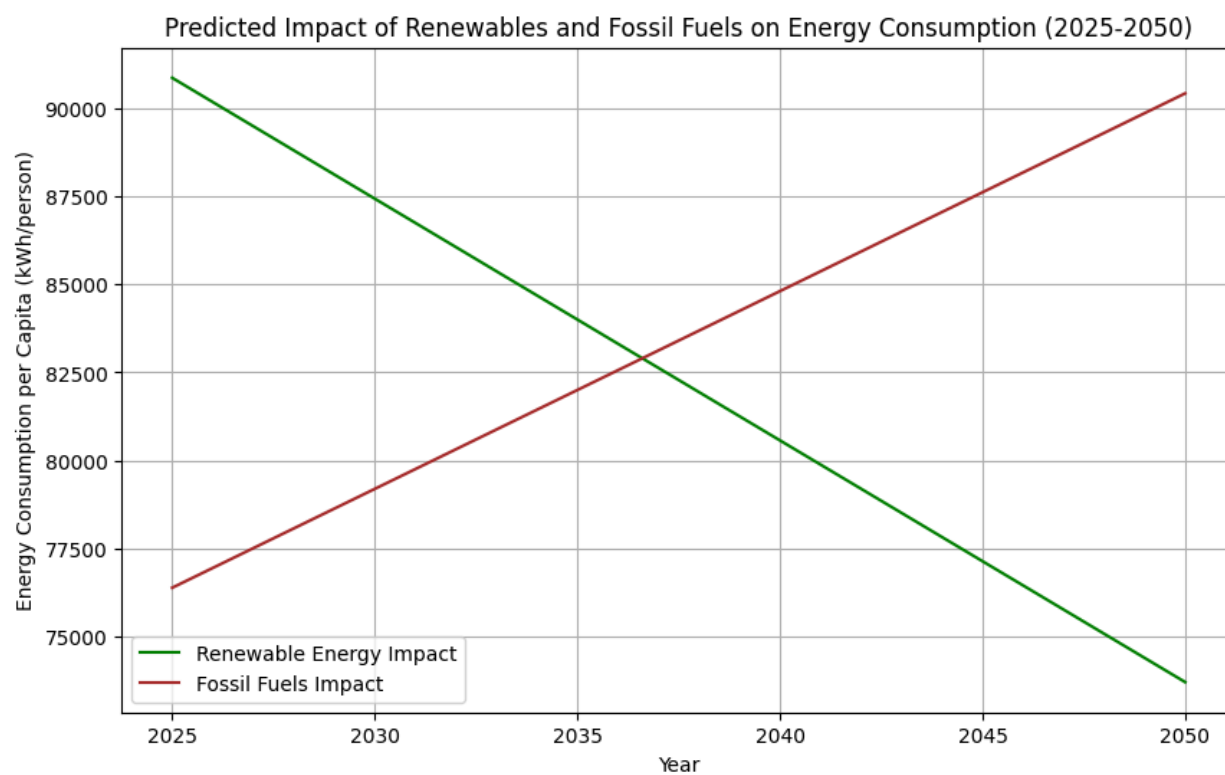


Figure 5. Prediction of Future Energy Consumption for Renewable Energy vs. Fossil Fuels



References

<https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks>